CL & NLP - LSTM Assignment

Question 1 — Single-cell LSTM forward pass

Given: $x_t = 0.5$, $h_{t-1} = 0.2$, $C_{t-1} = 0.4$. Sigmoid: $\sigma(z) = 1/(1 + e^{-z})$.

Step 1: Forget gate

$z_f = W_f x_t + U_f h_{t-1} + b_f = 0.7(0.5) + 0.3(0.2) + 0.1 = 0.3500 + 0.0600 + 0.1000 = 0.5100$

$f_t = \sigma(0.5100) = 1/(1 + e^{-0.5100}) = 0.6248$

Step 2: Input gate

$z_i = 0.6(0.5) + 0.4(0.2) + 0 = 0.3000 + 0.0800 = 0.3800$

$i_t = \sigma(0.3800) = 1/(1 + e^{-0.3800}) = 0.5939$

Step 3: Candidate cell state

$z_c = 0.9(0.5) + 0.2(0.2) + 0 = 0.4500 + 0.0400 = 0.4900$

$\tilde{C}_t = \tanh(0.4900) = (e^{0.4900} - e^{-0.4900})/(e^{0.4900} + e^{-0.4900}) = 0.4542$

Step 4: Updated cell state

$C_t = f_t C_{t-1} + i_t \tilde{C}_t = (0.6248 \times 0.4) + (0.5939 \times 0.4542)$

Forget contribution = 0.2499; input contribution = 0.2697 (each rounded to 4 decimal places).

$C_t = 0.2499 + 0.2697 = 0.5196$

Step 5: Output gate

$z_o = 0.5(0.5) + 0.5(0.2) + 0.1 = 0.2500 + 0.1000 + 0.1000 = 0.4500$

$o_t = \sigma(0.4500) = 0.6106$

Step 6: Hidden state

$h_t = o_t \times \tanh(C_t) = 0.6106 \times \tanh(0.5196)$

$\tanh(0.5196) = 0.4774$; therefore $h_t = 0.6106 \times 0.4774 = 0.2915$

Answer: $f_t = 0.6248$, $i_t = 0.5939$, $\tilde{C}_t = 0.4542$, $C_t = 0.5196$, $o_t = 0.6106$, $h_t = 0.2915$.

Question 2 — Number of LSTM trainable parameters

Given: input dimension d = 10; hidden dimension h = 32. Each of the four gates has input weights (h × d), recurrent weights (h × h), and biases (h).

(a) One bias vector per gate

Input weights per gate = h × d = 32 × 10 = 320.

Recurrent weights per gate = h × h = 32 × 32 = 1,024.

Biases per gate = h = 32.

Parameters per gate = 320 + 1,024 + 32 = 1,376.

Total for four gates = 4 × 1,376 = 5,504 parameters.

(b) Two bias vectors per gate (PyTorch convention)

Biases per gate = 2h = 2 × 32 = 64.

Parameters per gate = 320 + 1,024 + 64 = 1,408.

Total for four gates = 4 × 1,408 = 5,632 parameters.

Question 3 — Vanishing gradient through cell state

Gradient scaling over k time steps for a constant forget gate f is approximately f^k.

(a) f = 0.9

For k = 10: $0.9^{10} = 0.3486784401 \approx 0.3487$.

For k = 50: $0.9^{50} = 0.0051537752 \approx 0.0052$.

(b) k = 50, for two other forget-gate values

If f = 0.5: $0.5^{50} = 8.8818 \times 10^{-16}$ (≈ 0.0000 to four decimal places).

If f = 0.99: $0.99^{50} = 0.6050060671 \approx 0.6050$.

(c) Interpretation

A forget gate of 0.5 or 0.9 causes the gradient to shrink substantially over 50 steps, so learning a dependency that far back becomes difficult; with f = 0.99, about 60.50% of the gradient magnitude remains along this cell-state path. An LSTM can learn to keep its forget gate close to 1 and preserve information through an additive cell-state update, which reduces vanishing gradients compared with the repeatedly transformed hidden state of a vanilla RNN.

Question 4 — Electricity load forecasting

Given: 3 years, 365 days/year, 24 hours/day, lookback = 168 hours, prediction horizon = 24 hours, stride = 1, features = 3.

(a) Number of hourly records

N = 3 × 365 × 24 = 26,280 hourly records.

Answer: 26,280 records.

(b) Number of input/output windows

Each window uses 168 input hours followed by 24 target hours, covering 168 + 24 = 192 consecutive records.

Number of windows = ⌊(N − lookback − horizon)/stride⌋ + 1

= ⌊(26,280 − 168 − 24)/1⌋ + 1 = 26,088 + 1 = 26,089.

Answer: 26,089 windows.

(c) Tensor shapes for a batch of 64

Inputs: (batch size, time steps, features) = (64, 168, 3).

Outputs: (batch size, predicted hours) = (64, 24).

Answer: X = (64, 168, 3); y = (64, 24).

(d) Training windows and mini-batches

Training records = 70% × 26,280 = 0.70 × 26,280 = 18,396.

Training windows = 18,396 − 168 − 24 + 1 = 18,205.

Mini-batches per epoch = ⌈18,205/64⌉. Since 64 × 284 = 18,176, the remaining 29 windows form one final partial batch.

Answer: 18,205 training windows and 285 mini-batches (284 full + 1 partial).

(e) LSTM trainable parameters, single-bias convention

For input dimension d = 3, hidden size h = 64, and four gates: P = 4(hd + h² + h).

Input weights per gate = 64 × 3 = 192.

Recurrent weights per gate = 64 × 64 = 4,096.

Biases per gate = 64.

Parameters per gate = 192 + 4,096 + 64 = 4,352.

Answer: P = 4 × 4,352 = 17,408 trainable parameters.

Question 5 — Predictive maintenance

Given: minimum vibration = 20 mm/s, maximum vibration = 120 mm/s, reading = 95 mm/s; failure threshold = 0.5.

(a) Min-max normalisation

$x_{norm} = (x - x_{min})/(x_{max} - x_{min})$
$= (95 - 20)/(120 - 20) = 75/100 = 0.75.$

Answer: Normalised reading = 0.75.

(b) Binary prediction

Predicted probability = 0.83, threshold = 0.5. Since 0.83 ≥ 0.5, assign class 1 (failure).

Answer: The model predicts pump failure within the next cycle.

(c) Confusion matrix and performance metrics

Actual failures (positives) = 40; actual non-failures (negatives) = 1,000 − 40 = 960.

Predicted failures = 45; correct predicted failures = 30 ⇒ TP = 30.

False positives: FP = 45 − 30 = 15.

False negatives: FN = 40 − 30 = 10.

True negatives: TN = 960 − 15 = 945.

Check: TP + FP + FN + TN = 30 + 15 + 10 + 945 = 1,000.

Precision = TP/(TP + FP) = 30/(30 + 15) = 30/45 = 0.6667 = 66.67%.

Recall = TP/(TP + FN) = 30/(30 + 10) = 30/40 = 0.7500 = 75.00%.

F1-score = 2TP/(2TP + FP + FN) = 60/(60 + 15 + 10) = 60/85 = 0.7059 = 70.59%.

Accuracy = (TP + TN)/Total = (30 + 945)/1,000 = 975/1,000 = 0.9750 = 97.50%.

Answer: TP = 30, FP = 15, FN = 10, TN = 945; precision = 66.67%, recall = 75.00%, F1 = 70.59%, accuracy = 97.50%.

(d) Most important metric and the accuracy issue

The maintenance team should prioritise recall for the failure class, because missing an actual pump failure (a false negative) can cause unplanned downtime, equipment damage, or safety problems. Accuracy is misleading because only 40 of the 1,000 cycles are failures: a model predicting “no failure” for every cycle would achieve 960/1,000 = 96.00% accuracy but 0% failure recall. Precision should also be monitored, since excessive false alarms create unnecessary maintenance costs.